

SMART CODING TECHNIQUES And Their Benefits For Electronics



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Electronic devices feature a large number of small components, each piece being highly important to their overall functionality and safety. The responsibility for ensuring that each minute component is performing adequately rests firmly on the shoulders of the device and component manufacturers. However, their jobs are being made more difficult by the growing issue of counterfeiting in the electrical market.

Printed codes and marks enable manufacturers to track and trace products throughout the supply chain. These also, importantly, help distinguish between genuine and counterfeited components. However, counterfeiters are becoming increasingly sophisticated and have developed capabilities to replicate codes. In order to counter this problem, manufacturers find it necessary to employ more sophisticated coding methods such as smart code techniques.

Smart coding and micro-printing

In today's rapidly-changing electrical component industry, manufacturers must combine the right technology and techniques to encounter the many obstacles they face.

Electrical components, more often than not, appear to be identical. Coding and marking allows manufacturers to safely make a distinction between parts. Further, coding and marking allow them to build brand recognition into their products.

Application of increasingly-complex codes on small surface spaces is a challenge that is hard to overcome with outdated printer technologies. Advanced micro-printers have proved to be the solution. These have the ability to produce high-quality legible text down to 0.6mm in size in high-speed production environments.

Utilising algorithmic software technology, printers equipped with smart coding capabilities can alter and verify specific characters within codes. Implementation of smart coding techniques to improve existing basic lot/batch codes is simple, with nominal incremental production costs. Without knowledge of algorithms and keys, these smart codes are impossible to duplicate and can go a long way towards preventing counterfeiting activities, which contribute to huge profit losses.

Smart coding techniques

There are numerous smart coding techniques that component manufacturers can employ to enhance their basic lot/batch codes. Each technique (explained below) varies in complexity and application:

Self-verifying codes. These are one of the most basic ways to identify the authenticity of products with a visual check. With this type of marking, a pre-determined pattern is set for the code; for example, all digits add up to a certain number.

Interleaved marking. With interleaved marking, characters within a code are partially overlapped. While this overlay is visible to the naked eye, it

Fig. 1: 1650 and 1620 high-resolution CIJ printers from Videojet for enhanced micro-printing, at speeds up to 348 metres per minute

